Joint Optimization of User Scheduling, Rate Allocation, and Beamforming for RSMA Finite Blocklength Transmission

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***Abstract*—The forthcoming wireless network promises revolutionary advancements with significantly higher peak data rates, reduced latency, and vastly improved reliability. Among pivotal technologies, the design of novel multiple access schemes, particularly rate-splitting multiple access (RSMA), holds significant importance. In this article, we focus on the joint optimization of user scheduling, rate allocation, and beamforming for downlink multiple-input single-output communication networks under RSMA finite blocklength (FBL) transmission. The difficulty of the formulated optimization problem lies on the achievable rate function with FBL transmission and the joint design of user scheduling and beamforming. In order to solve the formulated problem, we first analyze the convexity and feasibility of the achievable rate function and further provide an efficient algorithm by cooperatively using strong Lagrangian duality, the difference of convex functions programming, the big-M method, and the alternating optimization algorithm for the joint optimization process. Numerical simulations validate the effectiveness of the proposed approach, offering promising insights for the future of 6G wireless networks.**

***Index Terms*—Beamforming design, finite blocklength (FBL) transmission, rate allocation, rate-splitting multiple access (RSMA), user scheduling.**

I. Introduction

IN CONTRAST to previous generations of wireless communication frameworks, the sixth generation (6G) wireless network will bring new features and enhancements to provide highly reliable and ubiquitous wireless connectivity. The 6G network is anticipated to attain a peak data rate 50 times greater, reduce latency by a factor of 10, and exhibit a reliability 10^4 times higher [1]. Additionally, it can also support a broader spectrum of core services, including enhanced mobile broadband (eMBB), ultra reliable low-latency communications (URLLCs), massive machine type communications (mMTCs), and some functional services, such as communications, sensing, localization, computing, and so on [2]. Consequently, innovative wireless resource management and anti-interference methods should be explored to address the challenges. As one of the key physical layer transmission technologies, the design of novel multiple access schemes holds significant importance [3].

In the conventional wireless communication system, space division multiple access (SDMA) was deemed as an effective multiple access scheme to boost system capacity [4]. However, the performance of exploiting the SDMA scheme is generally limited by the number of antennas at the transmitter [5]. To overcome the issue, the nonorthogonal multiple access (NOMA) technique was developed by utilizing superposition coding with successive interference cancelation (SIC), which can outperform the traditional scheme of orthogonal multiple access (OMA) [6], [7]. However, Clerckx et al. [8] initially analyzed the multiplexing gain of NOMA and proved that the performance improvement of NOMA will be limited in multiantenna settings, which is because users in each group are forced to completely decode messages from all co-scheduled users. Recently, rate-splitting multiple access (RSMA) has emerged as a novel framework in downlink multiuser multiantenna communication networks. Actually, the RSMA framework is able to adeptly integrate the two aforementioned multiple access methods, i.e., SDMA and NOMA. Specifically, RSMA splits user messages into common and private parts and then partially decodes the interference and treats the remaining part of the interference as noise, softly bridging the decoding principles of SDMA and NOMA. Moreover, RSMA also offers great advantages in terms of robustness, energy efficiency, spectrum efficiency, Quality of Service (QoS), and user fairness [9], [10], [11].

In the literature, a large amount of researches have investigated the performance and applications of RSMA. Mao et al. [5] developed the detailed RSMA framework

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and proved that RSMA scheme is superior to SDMA and NOMA both in network loads and user deployments. In [12], [13], and [14], the energy efficiency, the achievable sum rate, and the max-min fairness (MMF) were all studied in the framework of RSMA. Additionally, the RSMA scheme was also combined with other novel technologies. For example, Bansal et al. [15] and Katwe et al. [16] investigated the performance of RSMA for the intelligent reflecting surface (IRS)-aided system. Particularly Bansal et al. [15] derived a creative IRS-RSMA framework that leverages the interplay between rate-splitting and IRS. Xu et al. [17] and Cerna-Loli et al. [18] introduced RSMA to the dual-functional radar-communication transmission framework and found that RSMA not only effectively handles interference among communication users but also between communication and radar systems.

However, most existing works on RSMA considered the infinite blocklength transmission scenario. Indeed, with the emergence of increasingly latency-sensitive services, the research on the short-packet communication with finite blocklength (FBL) are imperative [19]. Considering the signal to interference plus noise ratio (SINR), the block error rate (BLER), and the blocklength, the achievable rate function of FBL transmission is more complicated than the conventional Shannon capacity [20]. Consequently, the complex rate function brings a great challenge to the optimization problem under FBL transmission scenarios [21]. To the best of our knowledge, only a few works investigated the RSMA scheme with FBL transmission. Specifically, Xu et al. [22] investigated the optimization of the sum rate for a two-user single-input single-output uplink system with FBL transmission. Moreover, the RSMA scheme with FBL transmission was applied in the multiple-antenna system and was proved to outperform schemes of the SDMA and the NOMA in terms of robustness, the MMF, and the achievable sum rate [14], [23], [24]. A flexible RSMA FBL scheme, where the system is capable of deciding whether SIC should be employed to decode the common stream or not, was innovatively proposed in [25]. Furthermore, RSMA for unmanned aerial vehicle (UAV)-assisted FBL communications was studied in [26], where the 3D-position of the UAV, power, and rate allocation are jointly optimized to achieve a higher ergodic sum rate.

As potential technologies of 6G, massive multiple-input–multiple-output (MIMO) and multiuser transmission are capable of meeting the stringent performance requirements [27]. Therefore, user scheduling and beamforming design will become two important issues in the upcoming wireless communication system. In fact, the joint design of user scheduling and beamforming is a mixed-discrete and continuous-variable programming problem and difficult to be solved directly. To tackle the difficulty, Dong et al. [28] designed an alternating iterative algorithm to jointly optimize user scheduling and beamforming aiming at maximizing the sum rate for integrated terrestrial-satellite networks. The alternating optimization method was also proposed in [29], where cross layer optimization was investigated with QoS support for joint transmission networks by designing user scheduling and beamforming alternately. In [30], a graph neural network-based joint user scheduling and beamforming learning algorithm was proposed to achieve close performance to the traditional method more efficiently. However, to the best of our knowledge, there are no existing works that investigate the joint design of user scheduling and beamforming for RSMA with FBL transmission.

In this article, considering the advantages and the necessity of RSMA and the FBL transmission, we focus on the joint optimization of user scheduling, rate allocation, and beamforming for the downlink multiuser multiple-input single-output (MISO) communication network under RSMA FBL transmission scenario. The main contributions are summarized as follows.

1) We consider the number of scheduled users as the performance metric for RSMA under FBL transmission. Specifically, for the multiuser MISO downlink system, the optimization problem is formulated to maximize the scheduled users set by jointly designing user scheduling, rate allocation, and beamforming with QoS and power budget constraints.
2) The convexity and monotonicity of the achievable rate function under FBL transmission are analyzed. The feasibility condition of the formulated problem is further provided, which simplifies the optimization process.
3) In order to solve the discrete-continuous variable optimization problem, we propose the efficient alternating optimization-based user scheduling, rate allocation, and beamforming (AO-based US-RA-BF) algorithm by cooperatively using the strong Lagrangian duality, the difference of convex functions (DC) programming, the big-M (BM) method, and the semidefinite relaxation (SDR).
4) Finally, the numerical results obtained by the AO-based US-RA-BF algorithm are evaluated via simulations, and the effectiveness of the proposed algorithm is proved.

The remainder of this article is organized as follows. Section II introduces the multiuser MISO RSMA FBL downlink system and formulates the joint optimization of user scheduling, rate allocation, and beamforming problem. Section III analyzes the rate function under FBL transmission and proposes the AO-based US-RA-BF algorithm to solve the formulated problem. Numerical results are shown in Section IV. Finally, Section V draws conclusions.

Notations: Bold lowercase and uppercase letters, such as $\mathbf{a}$ and $\mathbf{A}$, represent vectors and matrices, respectively. $|\cdot|$, $\|\cdot\|_0$, $\|\cdot\|_1$, and $\|\cdot\|_2$ indicate the absolute value, ℓ_0-norm, ℓ_1-norm, and ℓ_2-norm. $(\cdot)^{\mathrm{T}}$, $(\cdot)^{\mathrm{H}}$, and $(\cdot)^{-1}$ denote the transpose, conjugate transpose, and inverse of a matrix, respectively. $\mathrm{Rank}(\cdot)$, $\mathrm{tr}(\cdot)$, and $\mathbb{E}\{\cdot\}$ are, respectively, the rank, trace, and expectation of a matrix. $\mathbf{A} \succeq \mathbf{0}$ implies that $\mathbf{A}$ is a positive semidefinite matrix. $\mathbb{R}^{M\times 1}$, $\mathbb{R}_{+}^{M\times 1}$, and $\mathbb{C}^{M\times 1}$ are the M-dimensional real-valued, nonnegative real-valued, and complex-valued column vectors, respectively. $|\mathcal{A}|$ is the cardinality of the set $\mathcal{A}$. $\mathbf{I}_M$ is the $M\times M$ identity matrix.

II. System Model and Problem Formulation

A. System Model

As described in Fig. 1(a), we focus on a downlink communication network in which K single-antenna users are served by an N-antenna base station (BS). The K users are assumed to be randomly distributed in the range of (d_1, d_2) away from the BS and $d_1, d_2 \in (d_{\min}, d_{\max})$, where $d_{\min}$ and $d_{\max}$ denote the

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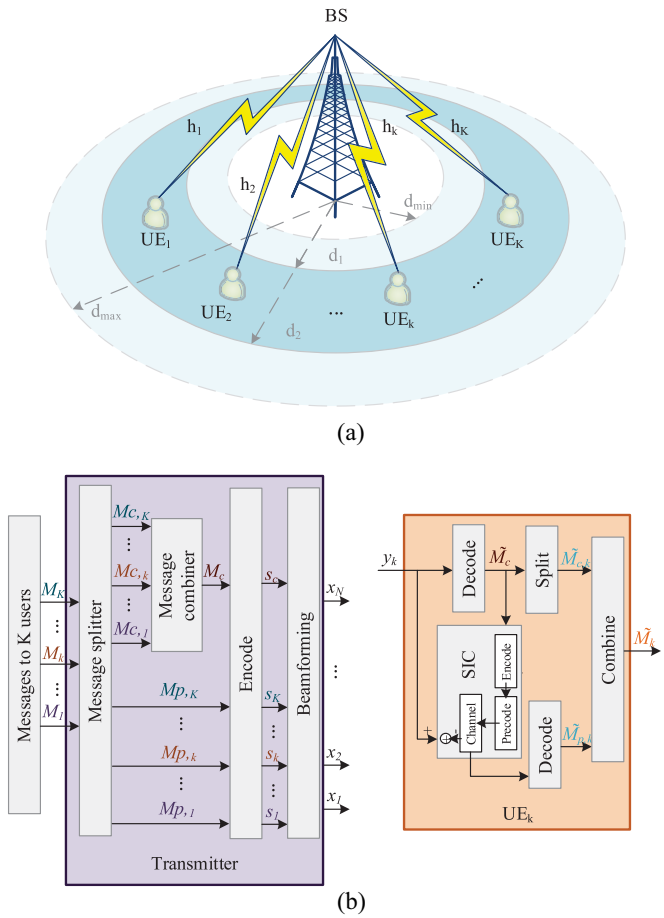

Fig. 1. Downlink MISO RSMA FBL system. (a) Distribution of users and (b) transceiver architecture of K users 1-layer RSMA.

reference distance and the cell radius, respectively [30]. Then, the scheduled users set $\mathcal{S} = \{1, 2, \ldots, S\}$ satisfying $\mathcal{S} \subseteq \mathcal{K}$ is selected from the candidate users set $\mathcal{K} = \{1, 2, \ldots, \mathrm{K}\}$. In this article, the 1-layer RSMA [8] scheme, which is shown in Fig. 1(b), is employed to facilitate the transmission. The message $M_k \quad \forall k \in \mathcal{K}$ sent to UE_k is divided into two parts, which are $M_{c,k}$ and $M_{p,k}$ denoting the common and private parts of the message, respectively. According to the RSMA principle [9], $\{M_{c,k}\}_{k=1}^{K}$ are encoded into a common stream s_c for all users to decode and each $M_{p,k}$ is encoded into the private stream s_k. Therefore, the overall transmission symbol stream vector is written as $\mathbf{s} = [s_c, s_1, s_2, \ldots, s_K]^T$, which is assumed to have unit power, i.e., $\mathbb{E}(\mathbf{s}\mathbf{s}^H) = \mathbf{I}$, and the BS transmits the signal

$$\mathbf{x} = \boldsymbol{w}_c s_c + \sum_{k \in \mathcal{K}} \boldsymbol{w}_k s_k. \tag{1}$$

In (1), $\boldsymbol{w}_c \in \mathbb{C}^{N\times 1}$ and $\boldsymbol{w}_k \in \mathbb{C}^{N\times 1}$ are the beamforming vectors encoding the common and private streams, respectively. The signal received at UE_k is

$$y_k = \mathbf{h}_k^{\mathrm{H}} \mathbf{x} + z_k = \mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_c s_c + \sum_{k \in \mathcal{K}} \mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_k s_k + z_k \tag{2}$$

where $\mathbf{h}_k \in \mathbb{C}^{N\times 1}$ denotes the channel between the BS and UE_k, and $z_k \sim \mathcal{CN}(0, \sigma_k^2)$ is the additive white Gaussian noise (AWGN) with zero mean and variance σ_k^2 at UE_k.

At each UE_k, SIC is employed. Specifically, UE_k first decodes the common stream s_c, treating all private stream interference as noise, and thus the SINR for decoding the common stream is written as

$$\Gamma_{c,k} = \frac{\left|\mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_c\right|^2}{\sum_{i \in \mathcal{K}} \left|\mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_i\right|^2 + \sigma_k^2}, k = 1, 2, \ldots, K. \tag{3}$$

Then, the common stream can be recovered from s_c under the shared codebook. With successful SIC, the common stream is removed from y_k and UE_k decodes the private stream s_i while treating the interference of the private streams from other users as noise. Thus, the SINR of the private stream decoded at UE_k is given by

$$\Gamma_k = \frac{\left|\mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_k\right|^2}{\sum_{i \in \mathcal{K}, i \neq k} \left|\mathbf{h}_k^{\mathrm{H}} \boldsymbol{w}_i\right|^2 + \sigma_k^2}, k = 1, 2, \ldots, K. \tag{4}$$

In order to meet the extreme requirements on delay, we focus on the FBL transmission in this work. Different from the traditional Shannon capacity, according to [20], the achievable rate for the FBL transmission is a function of the SINR, the BLER, and the blocklength. Accordingly, the common and private stream rates at UE_k, i.e., $R_{c,k}$ and R_k, are, respectively, given by

$$R_{c,k} = \log\left(1 + \Gamma_{c,k}\right) - \sqrt{\frac{V_{c,k}}{l_c}} Q^{-1}(\epsilon), k = 1, 2, \ldots, K \tag{5}$$

$$R_k = \log(1 + \Gamma_k) - \sqrt{\frac{V_k}{l_k}} Q^{-1}(\epsilon), k = 1, 2, \ldots, K \tag{6}$$

where l_c and $\{l_k\}_1^K$ represent the blocklengths of streams s_c and $\{s_k\}_1^K$ and ϵ is the BLER. The function $Q^{-1}(\cdot)$ is the inverse of Gaussian $\mathcal{Q}$ function, i.e., $Q(x) = (1/\sqrt{2\pi}) \int_x^\infty \exp\left[-(t^2/2)\right] \mathrm{d}t$. $V_{c,k}$, V_k are the channel dispersion, which are defined as

$$V_{c,k} = 1 - \left(1 + \Gamma_{c,k}\right)^{-2}, k = 1, 2, \ldots, K \tag{7}$$

$$V_k = 1 - (1 + \Gamma_k)^{-2}, k = 1, 2, \ldots, K. \tag{8}$$

B. Problem Formulation

This work focuses on maximizing the number of users in the scheduled users set $\mathcal{S}$ while satisfying QoS requirements and power budget constraint at the BS. Accordingly, the joint optimization of user scheduling, rate allocation, and beamforming problem is formulated as follows:

$$\max_{\mathcal{S}, \boldsymbol{w}_c, \{\boldsymbol{w}_k, r_k\}} \quad |\mathcal{S}| \tag{9a}$$

$$\text{s.t.} \quad \boldsymbol{w}_c^H \boldsymbol{w}_c + \sum_{k \in \mathcal{S}} \boldsymbol{w}_k^H \boldsymbol{w}_k \leq P_t \tag{9b}$$

$$\sum_{i \in \mathcal{S}} r_i \leq R_{c,k} \quad \forall k \in \mathcal{S} \tag{9c}$$

$$r_k + R_k \geq R_k^{\mathrm{th}} \quad \forall k \in \mathcal{S}. \tag{9d}$$

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In problem (9a), the objective function is denoted by $|\mathcal{S}|$, indicating the cardinality of $\mathcal{S}$. Constraint (9b) denotes the power budget constraint and P_t is the maximum transmission power at the BS. In (9c), each r_k is the common rate portion of the UE_k satisfying $\sum_{k\in\mathcal{S}} r_k = R_c$ and R_c is defined as $\min\{R_{c,1}, R_{c,2}, \ldots, R_{c,S}\}$, denoting the rate of the common stream, which enables all scheduled users to successfully decode the common stream. Constraint (9d) ensures QoS requirements for all scheduled users with the individual rate threshold $R_k^{\mathrm{th}} \quad \forall k \in \mathcal{S}$.

The formulated problem (9a) is a mixed-discrete and continuous-variable programming problem, which makes it difficult to find the global optimal solution. In this article, we propose the AO-based US-RA-BF algorithm to jointly optimize the user scheduling, rate allocation, and beamforming by exploiting the method of AO, which may be suboptimal but efficient [31], [32], [33]. Specifically, we first optimize the user scheduling and rate allocation with the given beamforming vectors. In this step, in order to optimize the user scheduling, the binary variables are introduced to represent the scheduled and unscheduled status of users and the problem is then transformed into a continuous type, which are further solved by employing strong Lagrangian duality, DC programming, and the BM method. Then, with given user scheduling and rate allocation, the beamforming vectors are further optimized by using SDR. Finally, the joint optimization of user scheduling, rate allocation, and beamforming algorithm, cooperatively using the DC programming, the BM, the SDR, and the AO algorithm, is proposed to solve the formulated optimization problem.

III. Joint Optimization of User Scheduling, Rate Allocation, and Beamforming

In this section, considering the difficulty of the rate constraints in the formulated optimization problem, we will first analyze the property of the rate function under FBL transmission. Then, strong Lagrangian duality, DC programming, and the BM method are cooperatively employed to optimize the user scheduling and rate allocation by fixing beamforming vectors, which are further optimized based on the property of rate function and the method of SDR.

A. Analysis of the Rate Function Under FBL Transmission

In this section, to jointly optimize the user scheduling, rate allocation, and beamforming, the property of the rate function for FBL transmission, i.e.,

$$R(\Gamma) = \log(1+\Gamma) - A\sqrt{V(\Gamma)} \tag{10}$$

where $A = ([Q^{-1}(\epsilon)]/\sqrt{l}) > 0$ due to $Q^{-1}(\epsilon) > 0$, is first analyzed. Note that work [21] pointed out that $R(\Gamma)$ always has an inflection point, which greatly affects the solution to the optimization problem (9a). Therefore, to dig out the convexity and monotonicity of the function $R(\Gamma)$, we equivalently transformed $R(\Gamma)$ into another form, i.e.,

$$R(\eta) = -\log\eta - A\sqrt{1-\eta^2} \tag{11}$$

with $\eta = [1/(1+\Gamma)] \in (0,1)$. In the following Proposition, the convexity and monotonicity of the function $R(\eta)$ are analyzed in detail.

Proposition 1: For $\eta \in (0,1)$, $R(\eta)$ is always a convex function.

Proof: See Appendix A. ■

Remark 1: However, according to (9c) and (9d) in the formulated problem (9a), the achievable rate function needs to be concave instead of a convex one. Therefore, the following Proposition further analyzes the monotonicity of the function $R(\eta)$.

Proposition 2: $R(\eta)$ decreases monotonically within $(0, \eta_0)$ and increases monotonically within $(\eta_0, 1)$; At the minimizer η_0, i.e.,

$$\eta_0 = \frac{\sqrt{\sqrt{1+4A^2}-1}}{\sqrt{2}A} \tag{12}$$

we always have $R(\eta_0) < 0$.

Proof: See Appendix B. ■

Remark 2: Using Proposition 2, it can be concluded that, in the range of $(0, \eta_0)$, there must exist a zero point of $R(\eta)$, i.e., $R(\eta_1) = 0$ satisfying $\eta_1 \in (0, \eta_0)$. However, (9c) and (9d) in the formulated problem (9a) are both in the form of $R(\eta^\star) \geq \alpha$ with α being a nonnegative constant. Therefore, the characterization of $\eta^\star$ is important to solve to problem (9a), which is described in the following Theorem.

Theorem 1: The analytical solution of $R(\eta) = \alpha (0 < \eta < 1, \alpha \geq 0)$ is

$$\eta^\star = \varphi(A,\alpha) = \mathrm{e}^{-\frac{\mathcal{W}\left(2A,-2A;-4A^2\mathrm{e}^{-2\alpha}\right)}{2}-\alpha} \tag{13}$$

where $\mathcal{W}$ is defined as

$$\mathcal{W}\left(\iota_1,\iota_2;\mu\right) = \iota_1 - \sum_{m=1}^{\infty}\frac{1}{m*m!}\left(\frac{\mu m\mathrm{e}^{-\iota_1}}{\iota_2-\iota_1}\right)^m \times \mathcal{B}_{m-1}\left(\frac{-2}{m(\iota_2-\iota_1)}\right) \tag{14}$$

with

$$\mathcal{B}_m(z) = \sum_{k=0}^{m}\frac{(m+k)!}{k!(m-k)!}\left(\frac{z}{2}\right)^k. \tag{15}$$

Proof: See Appendix C. ■

Using Theorem 1, the complex (9c) and (9d) can both be equivalently transformed into

$$\eta_{c,k} = \frac{1}{1+\Gamma_{c,k}} \leq \varphi\left(A_c, \sum_{i\in\mathcal{S}} r_i\right) \quad \forall k \in \mathcal{S} \tag{16}$$

$$\eta_k = \frac{1}{1+\Gamma_k} \leq \varphi\left(A_k, R_k^{\mathrm{th}} - r_k\right) \quad \forall k \in \mathcal{S} \tag{17}$$

with

$$A_c = \frac{Q^{-1}(\epsilon)}{\sqrt{l_c}} \tag{18}$$

$$A_k = \frac{Q^{-1}(\epsilon)}{\sqrt{l_k}}, k = 1, 2, \ldots, S. \tag{19}$$

In the following parts, based on the analysis of the rate function under FBL transmission, we further investigate the

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optimization of user scheduling, rate allocation, and beamforming.

B. User Scheduling Design and Rate Allocation

In this section, we focus on the user scheduling design and rate allocation with the given beamforming vectors. The difficulty of the problem lies on the uncertainty of the number of elements in $\mathcal{S}$. In order to obtain the optimized user scheduling and rate allocation results based on the formulated problem (9a), a binary variable x_k is introduced to represent the scheduled and unscheduled status of UE_k, i.e., $x_k = 1$ means that the UE_k is scheduled and vice versa is not, and hence the vector $\boldsymbol{x}= [x_1, x_2, \ldots, x_K]$ can release the uncertainty of the $\mathcal{S}$. According to problem (9a), the user scheduling and rate allocation problem can be equivalently transformed into

$$\max_{\{x_k, r_k\}} \quad \|\boldsymbol{x}\|_0 \tag{20a}$$
$$\text{s.t.} \quad x_k \in \{0, 1\} \quad \forall k \in \mathcal{K} \tag{20b}$$
$$x_k \sum_{i\in\mathcal{K}} r_i \le R_{c,k} \quad \forall k \in \mathcal{K} \tag{20c}$$
$$r_k + R_k \ge x_k R_k^{\mathrm{th}} \quad \forall k \in \mathcal{K} \tag{20d}$$

where $\boldsymbol{r}= [r_1, r_2, \ldots, r_{\mathrm{K}}]$ denotes the common rate allocation vector. In problem (20a), we use ℓ_0-norm of x to replace $|\mathcal{S}|$ in problem (9a). Note that some unscheduled users may still be assigned rates with small values. Therefore, (20c) and (20d) are stricter than (9c) and (9d), i.e., the solution to problem (9a) is the upper bound of problem (20a). One can easily find that the nonconvexity of problem (20a) lies on the objective function and (20c), which will be solved in the following.

First, in order to deal with (20c) in the form of a binary variable multiplied by a continuous variable, the subsequent Proposition presents the equivalent formulation of problem (20a).

Proposition 3: Using the BM method, y_k is introduced to transform (20c) into a convex linear form equivalently. Therefore, problem (20a) can be rewritten as

$$\max_{\{x_k, r_k, y_k\}} \quad \|\boldsymbol{x}\|_0 \tag{21a}$$
$$\text{s.t.} \quad y_k \le \min\left\{\sum_{i\in\mathcal{K}} r_i, R_{c,k} x_k\right\} \quad \forall k \in \mathcal{K} \tag{21b}$$
$$y_k \ge \max\left\{\sum_{i\in\mathcal{K}} r_i - R_{c,k}(1 - x_k), 0\right\} \quad \forall k \in \mathcal{K} \tag{21c}$$
$$\text{(20b), (20d).} \tag{21d}$$

Proof: See Appendix D. ■

However, problem (21a) is also a difficult mixed-discrete and continuous-variable programming problem. By converting the binary variable x_k to a continuous form, problem (21a) is equivalently written as

$$\min_{\{x_k, r_k, y_k\}} \quad -\sum_{k\in\mathcal{K}} x_k \tag{22a}$$
$$\text{s.t.} \quad 0 \le x_k \le 1 \quad \forall k \in \mathcal{K} \tag{22b}$$
$$\sum_{k\in\mathcal{K}} \left(x_k - x_k^2\right) \le 0 \tag{22c}$$
$$\text{(20d), (21b), (21c).} \tag{22d}$$

To deal with the nonconvex (22c) in problem (22a), the following Proposition gives the equivalent form of problem (22a).

Proposition 4: Using strong Lagrangian duality, problem (22a) is equivalently transformed into

$$\min_{\{x_k, r_k, y_k\}} \quad -\sum_{k\in\mathcal{K}} x_k + g(\boldsymbol{x}) \tag{23a}$$
$$\text{s.t.} \quad \text{(20d), (21b), (21c), (22b)} \tag{23b}$$

where $g(\boldsymbol{x})$ is defined as

$$g(\boldsymbol{x}) \triangleq \lambda \sum_{k\in\mathcal{K}} x_k - \lambda \sum_{k\in\mathcal{K}} x_k^2 \tag{24}$$

with $\lambda \in [0, 1]$ denoting the penalty factor.

Proof: See Appendix E. ■

Using Proposition 4, since the objective function in (23a) is the difference of two convex functions, DC programming is further employed [34]. Therefore, we introduce the following approximate form by using the successive convex approximation of $\lambda \sum_{k\in\mathcal{K}} x_k^2$:

$$h(\boldsymbol{x}) \triangleq \lambda \sum_{k\in\mathcal{K}} x_k - \lambda \left(\sum_{k\in\mathcal{K}} \left(x_k^{(t)}\right)^2 + 2 \sum_{k\in\mathcal{K}} x_k^{(t)} \left(x_k - x_k^{(t)}\right)\right) \tag{25}$$

where t denotes the tth iteration of the DC programming. Then, problem (23a) can be transformed into solving a set of convex approximation problems

$$\min_{\{x_k, r_k, y_k\}} \quad -\sum_{k\in\mathcal{K}} x_k + h(\boldsymbol{x}) \tag{26a}$$
$$\text{s.t.} \quad \text{(20d), (21b), (21c), (22b).} \tag{26b}$$

Algorithm 1 DC Programming

1: **Initialize** $t \leftarrow 0$, $\boldsymbol{x}^{(t)}, \boldsymbol{r}^{(t)}, \boldsymbol{y}^{(t)}, \zeta^{(t)}$.
2: **repeat**
3: $\quad t \leftarrow t + 1$.
4: $\quad$ Solve problem (26a) to obtain $\boldsymbol{x}^{(t)}, \boldsymbol{r}^{(t)}, \boldsymbol{y}^{(t)}$, and $\zeta^{(t)}$.
5: **until** $|\frac{\zeta^{(\mathrm{t})} - \zeta^{(\mathrm{t}-1)}}{\zeta^{(\mathrm{t}-1)}}| \le \delta$.
6: **Output** $\zeta^\star \leftarrow \zeta^{(t)}$, $\boldsymbol{x}^\star \leftarrow \boldsymbol{x}^{(t)}$, $\boldsymbol{r}^\star \leftarrow \boldsymbol{r}^{(t)}$, $\boldsymbol{y}^\star \leftarrow \boldsymbol{y}^{(t)}$.

The detailed DC programming steps are summarized in Algorithm 1. We first initialize variables of problem (26a) that satisfy (20d), (21b), (21c), and (22b), and $\zeta^{(t)}$ representing the value of the objective function in problem (26a). Then, problem (26a) is solved by using some convex tools, such as CVX, until the value of (26a) reaches convergence, i.e., $|([\zeta^{(\mathrm{t})} - \zeta^{(\mathrm{t}-1)}]/[\zeta^{(\mathrm{t}-1)}])| \le \delta$, with δ being the maximum permissible error.

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C. Beamforming Design

In this section, we further optimize the beamforming vectors with optimized user scheduling and rate allocation, and the corresponding beamforming problem is given by

$$\text{Find} \quad \boldsymbol{w}_c, \{\boldsymbol{w}_k\} \tag{27a}$$
$$\text{s.t.} \quad \boldsymbol{w}_c^H \boldsymbol{w}_c + \sum_{k\in\mathcal{K}} x_k \boldsymbol{w}_k^H \boldsymbol{w}_k \le P_t \tag{27b}$$
$$R_{c,k} \ge x_k \sum_{i\in\mathcal{K}} r_i \quad \forall k \in \mathcal{K} \tag{27c}$$
$$R_k \ge x_k R_k^{\text{th}} - r_k \quad \forall k \in \mathcal{K}. \tag{27d}$$

Problem (27a) is difficult to solve due to the nonconvex (27c) and (27d). However, using Proposition 2 and Theorem 1, problem (27a) can be equivalently transformed into

$$\text{Find} \quad \boldsymbol{w}_c, \{\boldsymbol{w}_k\} \tag{28a}$$
$$\text{s.t.} \quad \boldsymbol{w}_c^H \boldsymbol{w}_c + \sum_{k\in\mathcal{K}} x_k \boldsymbol{w}_k^H \boldsymbol{w}_k \le P_t \tag{28b}$$
$$\frac{1}{1+\Gamma_{c,k}} \le \varphi\left(A_c, x_k \sum_{i\in\mathcal{K}} r_i\right) \quad \forall k \in \mathcal{K} \tag{28c}$$
$$\frac{1}{1+\Gamma_k} \le \varphi\left(A_k, x_k R_k^{\text{th}} - r_k\right) \quad \forall k \in \mathcal{K} \tag{28d}$$

with

$$A_c = \frac{Q^{-1}(\epsilon)}{\sqrt{l_c}} \tag{29}$$
$$A_k = \frac{Q^{-1}(\epsilon)}{\sqrt{l_k}}, k = 1, 2, \ldots, K. \tag{30}$$

It is worth pointing out that (28c) and (28d) may cause the infeasibility of problem (9a). Therefore, we have the following Corollary.

Corollary 1: In order to guarantee that problem (9a) is feasible, we must have

$$\frac{1}{1+\Gamma_{c,k}} \le \varphi(A_c, 0) \quad \forall k \in \mathcal{K} \tag{31}$$
$$\frac{1}{1+\Gamma_k} \le \varphi(A_k, 0) \quad \forall k \in \mathcal{K}. \tag{32}$$

Proof: This can be easily proved based on the conclusions obtained in Proposition 2 and Theorem 1. ■

In Corollary 1, the proposed conditions actually guarantee that the rates of scheduled users decoding common and private streams are both nonnegative. Then, for ease of writing, for $k = 1, 2, \ldots, K$, we let

$$\eta_{c,k}^\star = \min\left\{\varphi\left(A_c, x_k \sum_{i\in\mathcal{K}} r_i\right), \varphi(A_c, 0)\right\} \tag{33}$$
$$\eta_k^\star = \min\left\{\varphi\left(A_k, x_k R_k^{\text{th}} - r_k\right), \varphi(A_k, 0)\right\}. \tag{34}$$

Thus, problem (28a) is further reformulated as

$$\text{Find} \quad \boldsymbol{w}_c, \{\boldsymbol{w}_k\} \tag{35a}$$
$$\text{s.t.} \quad \boldsymbol{w}_c^H \boldsymbol{w}_c + \sum_{k\in\mathcal{K}} x_k \boldsymbol{w}_k^H \boldsymbol{w}_k \le P_t \tag{35b}$$
$$\Gamma_{c,k} \ge \frac{1-\eta_{c,k}^\star}{\eta_{c,k}^\star} \quad \forall k \in \mathcal{K} \tag{35c}$$
$$\Gamma_k \ge \frac{1-\eta_k^\star}{\eta_k^\star} \quad \forall k \in \mathcal{K}. \tag{35d}$$

In order to solve problem (35a), we introduce $\boldsymbol{Q}_c = \boldsymbol{w}_c \boldsymbol{w}_c^H$ and $\{\boldsymbol{Q}_k\}_{k=1}^K = \{\boldsymbol{w}_k \boldsymbol{w}_k^H\}_{k=1}^K$. Then problem (35a) is rewritten as

$$\text{Find} \quad \boldsymbol{Q}_c \succeq \boldsymbol{0}, \{\boldsymbol{Q}_k \succeq \boldsymbol{0}\} \tag{36a}$$
$$\text{s.t.} \quad \text{tr}(\boldsymbol{Q}_c) + \sum_{k\in\mathcal{K}} x_k \text{tr}(\boldsymbol{Q}_k) \le P_t \tag{36b}$$
$$\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_c \mathbf{h}_k - \frac{1-\eta_{c,k}^\star}{\eta_{c,k}^\star} \times \left(\sum_{i\in\mathcal{K}} (\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_i \mathbf{h}_k) + \sigma_k^2\right) \ge 0 \quad \forall k \in \mathcal{K} \tag{36c}$$
$$\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_k \mathbf{h}_k - \frac{1-\eta_k^\star}{\eta_k^\star} \times \left(\sum_{i\in\mathcal{K}, i\ne k} (\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_i \mathbf{h}_k) + \sigma_k^2\right) \ge 0 \quad \forall k \in \mathcal{K} \tag{36d}$$
$$\text{Rank}(\boldsymbol{Q}_c) \le 1 \tag{36e}$$
$$\text{Rank}(\boldsymbol{Q}_k) \le 1 \quad \forall k \in K. \tag{36f}$$

The nonconvexity of problem (36a) only lies on the rank (36e) and (36f). Therefore, the SDR method is adopted and we first solve the problem without rank constraints, i.e.,

$$\text{Find} \quad \boldsymbol{Q}_c \succeq \boldsymbol{0}, \{\boldsymbol{Q}_k \succeq \boldsymbol{0}\} \tag{37a}$$
$$\text{s.t.} \quad \text{tr}(\boldsymbol{Q}_c) + \sum_{k\in\mathcal{K}} x_k \text{tr}(\boldsymbol{Q}_k) \le P_t \tag{37b}$$
$$\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_c \mathbf{h}_k - \frac{1-\eta_{c,k}^\star}{\eta_{c,k}^\star} \times \left(\sum_{i\in\mathcal{K}} (\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_i \mathbf{h}_k) + \sigma_k^2\right) \ge 0 \quad \forall k \in \mathcal{K} \tag{37c}$$
$$\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_k \mathbf{h}_k - \frac{1-\eta_k^\star}{\eta_k^\star} \times \left(\sum_{i\in\mathcal{K}, i\ne k} (\mathbf{h}_k^{\text{H}} \boldsymbol{Q}_i \mathbf{h}_k) + \sigma_k^2\right) \ge 0 \quad \forall k \in \mathcal{K}. \tag{37d}$$

Note that problem (37a) is a standard semidefinite programming (SDP), so it can be solved efficiently by using convex optimization tools, such as CVX. However, problems (36a) and (37a) are not strictly equivalent, since the solutions to problem (37a) may not always satisfy the rank (36e) and (36f). Therefore, we will use the singular value decomposition (SVD) of $\boldsymbol{Q}_c$ and $\{\boldsymbol{Q}_k\}_{k=1}^K$ which are produced by using randomization technique [35] to obtain the beamforming vectors if the solutions to problem (37a) are not rank-one matrices [36].

Finally, we propose the AO-based US-RA-BF algorithm to alternately optimize the user scheduling, rate allocation, and beamforming. The detailed AO-based US-RA-BF algorithm steps are summarized in Algorithm 2. In Algorithm 2, we first

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Algorithm 2 AO-Based US-RA-BF Algorithm

Initialize $\tau \leftarrow 0$, $\boldsymbol{w_c}^{(\tau)}$, $\{\boldsymbol{w_k}^{(\tau)}\}_{k=1}^{K}$, and $\zeta^{(\tau)}$.
repeat
  $\tau \leftarrow \tau + 1$.
  Solve problem (26a) using Algorithm 1, $\boldsymbol{w_c}^{(\tau-1)}$, and $\{\boldsymbol{w_k}^{(\tau-1)}\}_{k=1}^{K}$ to obtain $\boldsymbol{x}^{(\tau)}$, $\boldsymbol{r}^{(\tau)}$, $\boldsymbol{y}^{(\tau)}$, and $\zeta^{(\tau)}$.
  Solve problem (37a) using $\boldsymbol{x}^{(\tau)}, \boldsymbol{r}^{(\tau)}, \boldsymbol{y}^{(\tau)}$, and $\zeta^{(\tau)}$ to obtain $\boldsymbol{w_c}^{(\tau)}$, $\{\boldsymbol{w_k}^{(\tau)}\}_{k=1}^{K}$.
until $|\frac{\zeta^{(\tau)}-\zeta^{(\tau-1)}}{\zeta^{(\tau-1)}}| \leq \xi$.
Output $\zeta^{\star} \leftarrow \zeta^{(\tau)}$, $\boldsymbol{x}^{\star} \leftarrow \boldsymbol{x}^{(\tau)}$, $\boldsymbol{r}^{\star} \leftarrow \boldsymbol{r}^{(\tau)}$, $\boldsymbol{y}^{\star} \leftarrow \boldsymbol{y}^{(\tau)}$, $\boldsymbol{w_c}^{\star} \leftarrow \boldsymbol{w_c}^{(\tau)}$, $\{\boldsymbol{w_k}^{\star}\}_{k=1}^{K} \leftarrow \{\boldsymbol{w_k}^{(\tau)}\}_{k=1}^{K}$.

TABLE I
TABLE OF PARAMETERS

BS antenna number N	16
AWGN power σ_k^2	0.0001
Path loss exponent υ	3
BLER ϵ	10^{-6}
Reference distance and cell radius $(d_{\min}, d_{\max})$	(50m, 300m)
User distribution range (d_1, d_2)	(200m, 250m)

initialize $\boldsymbol{w_c}^{(\tau)}$ and $\{\boldsymbol{w_k}^{(\tau)}\}_{k=1}^{K}$ that satisfy (37b), and $\zeta^{(\tau)}$ representing the value of the objective function in problem (26a), where τ denotes the τth iteration of Algorithm 2. Then, problems (26a) and (37a) will be solved alternately by using each other's previous solutions, where problem (26a) also uses Algorithm 1, until the value of (26a) reaches convergence, i.e., $|([\zeta^{(\tau)} - \zeta^{(\tau-1)}]/[\zeta^{(\tau-1)}])| \leq \xi$, with ξ being the maximum permissible error in Algorithm 2.

IV. Simulation Results

In this section, we present the numerical results achieved by our proposed AO-based US-RA-BF algorithm for the joint optimization of user scheduling, rate allocation, and beamforming in the downlink multiuser MISO RSMA FBL transmission system. In our simulation, the channel of UE_k is defined as $\mathbf{h}_k = \sqrt{\rho_k}\mathbf{g}_k \in \mathbb{C}^{N\times 1}$, where $\mathbf{g}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ is the small-scale fading and $\rho_k = 1/(1 + (d_k/d_{\min})^{\upsilon})$ denotes the path-loss with $d_k \in (d_1, d_2)$. We assume the same noise power for all users. Throughout our simulations, the performance in terms of the number of scheduled users and sum rate is evaluated by considering the change in power budget of the BS P_t, the rate threshold R_k^{th}, the number of users K, and the blocklength $l_k, k = c, 1, \ldots, K$. The simulation parameters are summarized in Table I

Fig. 2 illustrates the number of scheduled users versus the rate threshold with different power budgets of the BS, i.e., $P_t = 5$ W, $P_t = 10$ W, $P_t = 15$ W, and $P_t = 20$ W. As expected, with the increasing the rate threshold, fewer and fewer users will be selected from the user set $\mathcal{K}$. Additionally, it is easy to find that more users will be scheduled by increasing the power budget of the BS, which is because the rise in power contributes to better SINR, enabling a greater number of users to fulfill QoS requirements.

In Fig. 3, we show the number of scheduled users versus the blocklength, also with different power budgets of the BS

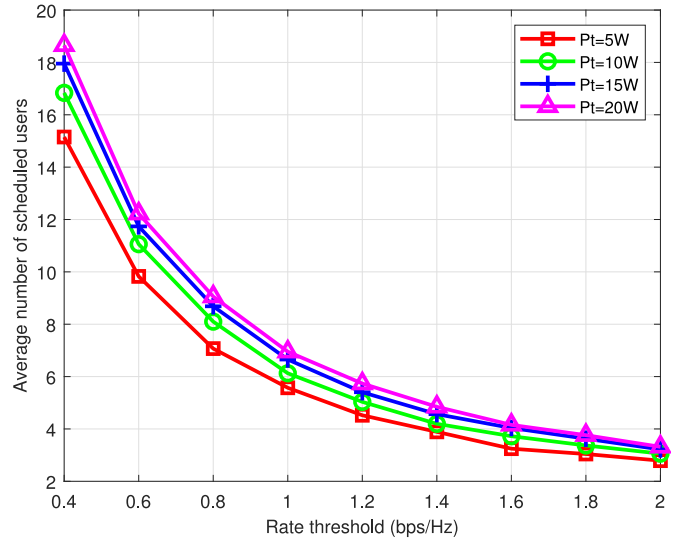

Fig. 2. Number of scheduled users versus the rate threshold. We set $K = 20$ and $l_k = 1024$ bit, $k = c, 1, \ldots, K$.

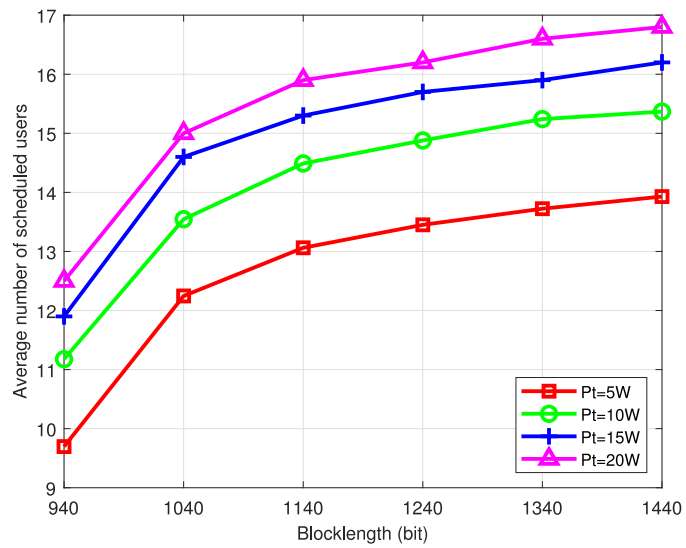

Fig. 3. Number of scheduled users versus the blocklength. We set $K = 20$ and $\{R_k\text{th}\}_{k=1}^{K} = 0.5$ bps/Hz.

as Fig. 2. Note that longer blocklength induces higher rate and the users are easier to satisfy the QoS requirements. Therefore, we can find that the number of scheduled users is improved by increasing blocklength. Furthermore, when the blocklength is large enough, the joint user scheduling, rate allocation, and beamforming design based on FBL transmission converges toward a scenario resembling infinite blocklength transmission.

Fig. 4 depicts the proportion of scheduled users (number of scheduled users/total number of users) versus the total number of users under the same power conditions in Figs. 2 and 3. As seen from the figure, the proportion of scheduled users decreases by increasing the total number of users. This phenomenon arises due to increased competition among users for finite resources, thereby restricting the pool of users who can effectively fulfill QoS requirements. In

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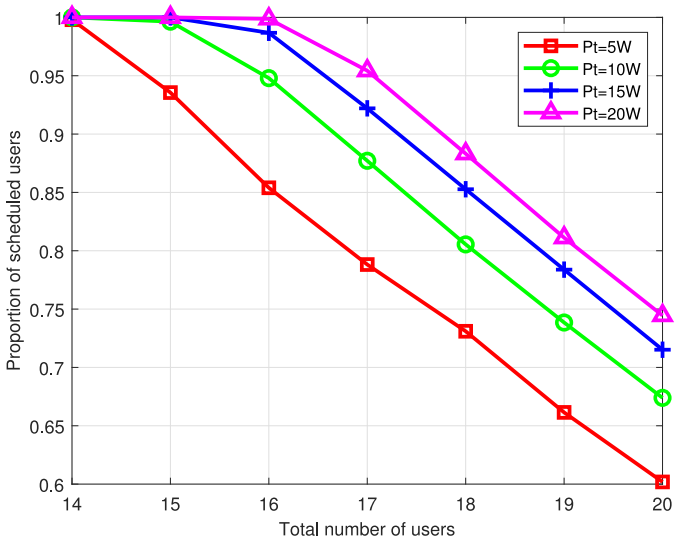

Fig. 4. Proportion of scheduled users versus the total number of users. We set $l_k = 1024$ bit, $k = c, 1, \ldots, K$ and $\{R_k\text{th}\}_{k=1}^{K} = 0.5$ bps/Hz.

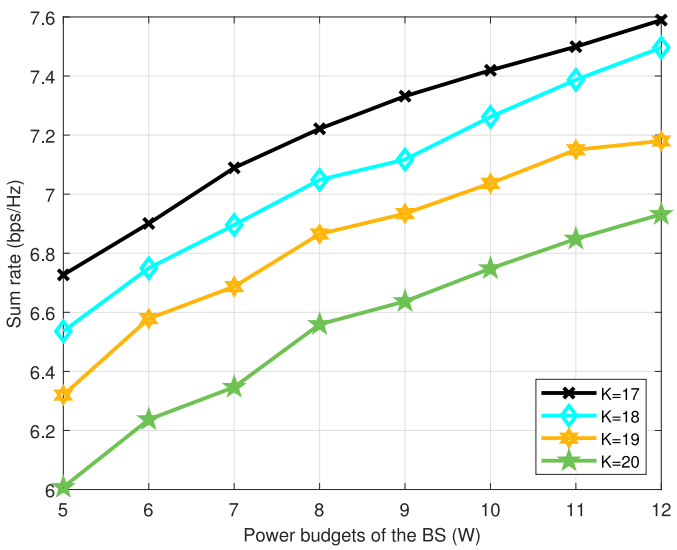

Fig. 5. Sum rate versus the power budgets of the BS. We set $l_k = 1024$ bit, $k = c, 1, \ldots, K$ and $\{R_k\text{th}\}_{k=1}^{K} = 0.5$ bps/Hz.

addition, similar to Figs. 2 and 3, the proportion of scheduled users will improve, by increasing the power budgets of the BS.

Fig. 5 displays the sum rate R_{sum}, which is computed by $R_{\text{sum}} = \sum_{k=1}^{K}(r_k + R_k)$, versus the power budgets of the BS with different total numbers of users, i.e., $K = 14$, $K = 16$, $K = 18$, and $K = 20$. Similar to Fig. 4, one can find that the sum rate decreases with a larger K. This can be attributed to the augmented overall user count, resulting in a decrease in the number of users being scheduled. Furthermore, in Fig. 6, we compare the performance of our RSMA scheme with the traditional OMA scheme. It is obvious that the sum rate obtained by the RSMA scheme is superior to that of the OMA scheme, particularly in the scenario with a larger total number of users. This is because, in contrast to the OMA scheme, where users individually occupy communication resources, users using RSMA scheme will share the limited resources.

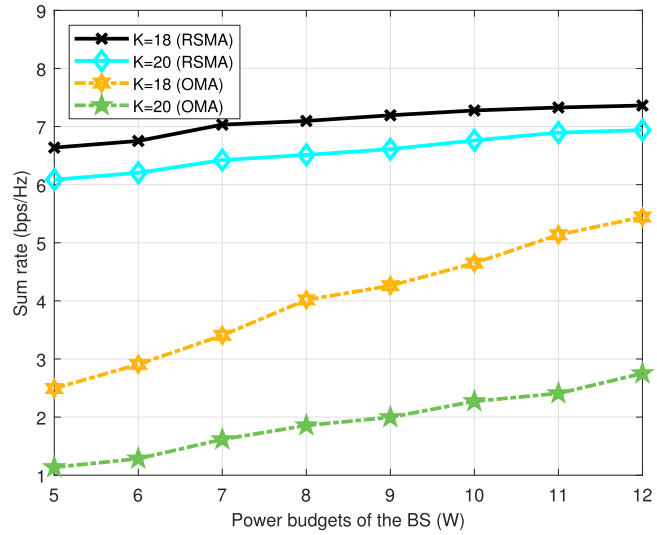

Fig. 6. Sum rate versus the power budgets of the BS. We set $l_k = 1024$ bit, $k = c, 1, \ldots, K$ and $\{R_k\text{th}\}_{k=1}^{K} = 0.4$ bps/Hz.

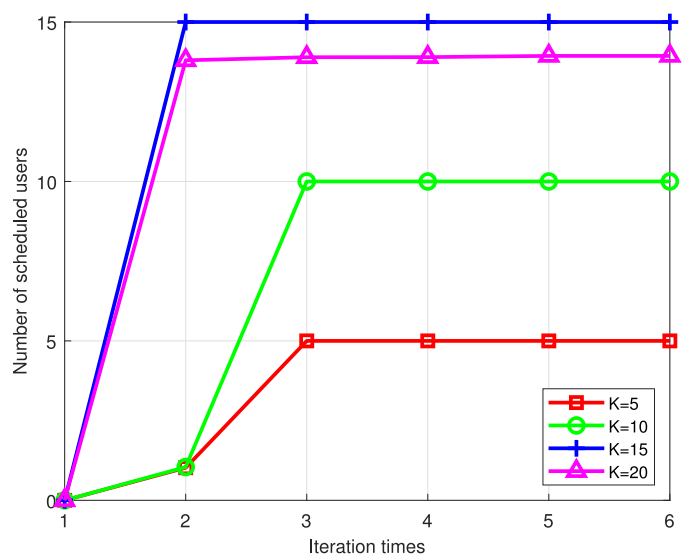

Fig. 7. Convergence of the AO-based US-RA-BF algorithm.

Finally, Fig. 7 shows the convergence of the AO-based US-RA-BF algorithm. In Fig. 7, the proposed algorithm always takes a few iterations to reach the optimal solutions with different total number of users, which implies the convergence of our proposed algorithm.

V. Conclusion

In this work, we study the joint optimization of user scheduling, rate allocation, and beamforming in a downlink multiuser MISO RSMA FBL transmission system aiming at maximizing the number of scheduled users considering the QoS and the power budget constraints. To address the formulated problem, we first focus on the analysis of the achievable rate function under FBL transmission and prove the convexity and monotonicity of the rate function. Then, the joint optimization process is solved by using the AO method, i.e., we first optimize the user scheduling and rate allocation by using strong Lagrangian duality, DC programming, and

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the BM method with given beamforming, which is further optimized by using SDR. Finally, an efficient AO-based US-RA-BF algorithm for the joint design of user scheduling, rate allocation, and beamforming is proposed. Simulation results are provided to prove the effectiveness of the AO-based US-RA-BF algorithm.

Appendix A
Proof of Proposition 1

From (11), the second order derivation of function $R(\eta)$ is given by

$$R''(\eta) = \frac{A}{\left(1-\eta^2\right)^{\frac{3}{2}}} + \frac{1}{\eta^2}. \tag{38}$$

Since $\eta = [1/(1+\Gamma)] \in (0,1)$ and $A > 0$, we always have $R''(\eta) > 0$, implying the convexity of function $R(\eta)$.

Appendix B
Proof of Proposition 2

The first order derivation of function $R(\eta)$ is given by

$$R'(\eta) = \frac{A\eta^2 - \sqrt{1-\eta^2}}{\eta\sqrt{1-\eta^2}}. \tag{39}$$

Therefore, we can obtain the stationary point of $R(\eta)$ as

$$\eta_0 = \frac{\sqrt{\sqrt{1+4A^2}-1}}{\sqrt{2}A}. \tag{40}$$

Note that the function $A\eta^2 - \sqrt{1-\eta^2}$ is a monotonic increasing function within $(0,1)$. Thus, $R(\eta)$ monotonically decreases for $0 < \eta < \eta_0$ and monotonically increases for $\eta_0 < \eta < 1$, and $R(\eta_0) < 0$ due to $R(1) = 0$.

Appendix C
Proof of Theorem 1

First, $R(\eta) = \alpha$ is rewritten as

$$\eta = \mathrm{e}^{-A\sqrt{1-\eta^2}-\alpha}. \tag{41}$$

Let $k = A\sqrt{1-\eta^2} \in (0, A)$, implying $\eta = [(\sqrt{A^2-k^2})/A]$, and (41) is equivalently transformed into $\eta = \mathrm{e}^{-k-\alpha}$. Hence, we obtain

$$\frac{\sqrt{A^2-k^2}}{A} = \mathrm{e}^{-k-\alpha}. \tag{42}$$

After some basic mathematical operations, (42) can be rewritten as

$$\mathrm{e}^{2k}(A+k)(A-k) = A^2\mathrm{e}^{-2\alpha}. \tag{43}$$

It is worth pointing out that (43) is actually the form of a transcendent equation (generalized Lambert W function) [37]. In order to solve (43), let $\beta = 2k \in (0, 2A)$ and (43) can be reformulated as

$$\mathrm{e}^{\beta}(\beta+2A)(\beta-2A) = -4A^2\mathrm{e}^{-2\alpha}. \tag{44}$$

Within the range $0 < \beta < 2A$, we obtain the solution to (44) by using Lagrange inversion theorem [21]

$$\beta^\star = \mathcal{W}\left(^{2A,-2A}; -4A^2\mathrm{e}^{-2\alpha}\right) \tag{45}$$

and $\mathcal{W}$ is defined as

$$\mathcal{W}\left(^{\iota_1,\iota_2}; \mu\right) = \iota_1 - \sum_{m=1}^{\infty} \frac{1}{m*m!}\left(\frac{\mu m\mathrm{e}^{-\iota_1}}{\iota_2-\iota_1}\right)^m \times \mathcal{B}_{m-1}\left(\frac{-2}{m(\iota_2-\iota_1)}\right) \tag{46}$$

with

$$\mathcal{B}_m(z) = \sum_{k=0}^{m} \frac{(m+k)!}{k!(m-k)!}\left(\frac{z}{2}\right)^k. \tag{47}$$

Therefore, using $\eta = \mathrm{e}^{-k-\alpha}$, the solution to $R(\eta) = \alpha$ can be obtained as

$$\eta^\star = \varphi(A,\alpha) = \mathrm{e}^{-k^\star-\alpha} = \mathrm{e}^{-\frac{\beta^\star}{2}-\alpha} = \mathrm{e}^{-\frac{\mathcal{W}\left(^{2A,-2A}; -4A^2\mathrm{e}^{-2\alpha}\right)}{2}-\alpha}. \tag{48}$$

Appendix D
Proof of Proposition 3

First, introducing

$$y_k = x_k \sum_{i\in\mathcal{K}} r_i, k = 1, 2, \ldots, K, \tag{49}$$

constraint (20c) can be equivalently transformed into the following two constraints:

$$y_k = x_k \sum_{i\in\mathcal{K}} r_i \quad \forall k \in \mathcal{K} \tag{50}$$

$$y_k \le R_{c,k} \quad \forall k \in \mathcal{K}. \tag{51}$$

According to [38], (50) can be approximated by

$$y_k \le \sum_{i\in\mathcal{K}} r_i, y_k \le M_k x_k \quad \forall k \in \mathcal{K} \tag{52}$$

$$y_k \ge \sum_{i\in\mathcal{K}} r_i - M_k(1-x_k), y_k \ge 0 \quad \forall k \in \mathcal{K} \tag{53}$$

where $\{M_k\}_{k=1}^K$ are all significant constants. Furthermore, since (51) precisely defines the upper bound of $\{y_k\}_{k=1}^K$, we have $\{M_k\}_{k=1}^K = \{R_{c,k}\}_{k=1}^K$ and (52) and (53) can be transformed into

$$y_k \le \sum_{i\in\mathcal{K}} r_i, y_k \le R_{c,k}x_k \quad \forall k \in \mathcal{K} \tag{54}$$

$$y_k \ge \sum_{i\in\mathcal{K}} r_i - R_{c,k}(1-x_k), y_k \ge 0 \quad \forall k \in \mathcal{K}. \tag{55}$$

Noting that $x_k \in \{0,1\} \quad \forall k \in \mathcal{K}$, we further prove the equivalence between (20c) and (54) and (55) under two cases, i.e., $x_k = 1$ or $x_k = 0$. For the first case, if $x_k = 1$, (54) and (55) can be rewritten as

$$y_k \le \sum_{i\in\mathcal{K}} r_i, y_k \le R_{c,k} \tag{56}$$

$$y_k \ge \sum_{i\in\mathcal{K}} r_i, y_k \ge 0 \tag{57}$$

implying $y_k = \sum_{i\in\mathcal{K}} r_i \le R_{c,k}$. For the second case, $x_k = 0$, (54) and (55) are equivalent to the following constraints:

$$y_k \le \sum_{i\in\mathcal{K}} r_i, y_k \le 0 \tag{58}$$

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$$y_k \geq \sum_{i\in\mathcal{K}} r_i - R_{c,k},\ y_k \geq 0 \tag{59}$$

which indicate $y_k = 0$ due to $\sum_{i\in\mathcal{K}} r_i \leq R_{c,k}$. Therefore, (54) and (55) are equivalent to (20c). Further, it is easy to find that (54) and (55) are equivalent to

$$y_k \leq \min\left\{\sum_{i\in\mathcal{K}} r_i, R_{c,k}x_k\right\} \quad \forall k \in \mathcal{K} \tag{60}$$

$$y_k \geq \max\left\{\sum_{i\in\mathcal{K}} r_i - R_{c,k}(1-x_k), 0\right\} \quad \forall k \in \mathcal{K} \tag{61}$$

which completes the proof.

Appendix E
Proof of Proposition 4

To deal with the nonconvex (22c), we introduce the Lagrangian $\mathcal{L}(\boldsymbol{x},\lambda) = -\sum_{k\in\mathcal{K}} x_k + \lambda\sum_{k\in\mathcal{K}} x_k - \lambda\sum_{k\in\mathcal{K}} x_k^2$ with Lagrangian multiplier λ. Define the convex set $\mathcal{G}$ to be the feasible domain formed by constraints (20d), (21b), (21c), and (22b) in problem (22a), i.e.,

$$\mathcal{G} = \{(x_k, r_k, y_k)|(20d), (21b), (21c), (22b)\}. \tag{62}$$

Then, problem (22a) is rewritten as

$$\min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \max_{\lambda\geq 0} \mathcal{L}(\boldsymbol{x},\lambda) \tag{63}$$

whose Lagrangian duality is

$$\sup_{\lambda\geq 0} \min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \mathcal{L}(\boldsymbol{x},\lambda). \tag{64}$$

Assuming $\phi(\lambda)$ and $(\boldsymbol{x}^\lambda, \boldsymbol{r}^\lambda, \boldsymbol{y}^\lambda)$ are, respectively, the optimal objective function value of problem (23a) and solution to problem (23a), we have

$$\phi(\lambda) = \mathcal{L}\left(\boldsymbol{x}^\lambda, \lambda\right) = \min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \mathcal{L}(\boldsymbol{x},\lambda) \tag{65}$$

i.e.,

$$\sup_{\lambda\geq 0} \phi(\lambda) = \sup_{\lambda\geq 0} \min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \mathcal{L}(\boldsymbol{x},\lambda). \tag{66}$$

Denote $P^\star$ to be the optimal value of the problem (22a). According to [40], we obtain

$$\sup_{\lambda\geq 0} \min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \mathcal{L}(\boldsymbol{x},\lambda) \leq \min_{\{x_k,r_k,y_k\}\in\mathcal{G}} \max_{\lambda\geq 0} \mathcal{L}(\boldsymbol{x},\lambda) = P^\star \tag{67}$$

implying the following conclusion:

$$\sup_{\lambda\geq 0} \phi(\lambda) \leq P^\star. \tag{68}$$

Note that, due to the fact of $\sum_{k\in\mathcal{K}} x_k - \sum_{k\in\mathcal{K}} x_k^2 \geq 0$ for $0 \leq x_k \leq 1 \quad \forall k \in \mathcal{K}$, $\phi(\lambda)$ is monotonic increasing in λ and bounded by $P^\star$. Furthermore, since the property of $\phi(\lambda)$ would be different under $\sum_{k\in\mathcal{K}} x_k - \sum_{k\in\mathcal{K}} x_k^2 = 0$ and $\sum_{k\in\mathcal{K}} x_k - \sum_{k\in\mathcal{K}} x_k^2 > 0$, we will prove the strong Lagrangian duality for problem (22a) under the following two cases.

For the first case, suppose there exists $0 < \lambda_0^1 < \infty$ such that $\sum_{k\in\mathcal{K}} x_k^{(\lambda_0^1)} - \sum_{k\in\mathcal{K}} (x_k^{(\lambda_0^1)})^2 = 0$, then the solution of $(\boldsymbol{x}^{\lambda_0^1}, \boldsymbol{r}^{\lambda_0^1}, \boldsymbol{y}^{\lambda_0^1})$ is feasible to problem (22a), i.e.,

$$\phi\left(\lambda_0^1\right) = \mathcal{L}\left(\boldsymbol{x}^{\lambda_0^1}, \lambda_0^1\right) \geq P^\star. \tag{69}$$

Jointly considering (68), (69), and the fact that $\phi(\lambda)$ is a monotonic increasing function, we conclude that

$$\phi(\lambda) = P^\star \quad \forall \lambda \geq \lambda_0^1 \tag{70}$$

and $(\boldsymbol{x}^{\lambda_0^1}, \boldsymbol{r}^{\lambda_0^1}, \boldsymbol{y}^{\lambda_0^1})$ is the optimal solution to problems (22a) and (23a). Therefore, the strong Lagrangian duality holds for problem (22a), i.e., problem (22a) is equivalent to problem (23a) in this case.

For the second case, i.e., $\sum_{k\in\mathcal{K}} x_k^{(\lambda)} - \sum_{k\in\mathcal{K}} (x_k^{(\lambda)})^2 > 0$ for all $\lambda \geq 0$, supposing that $\sum_{k\in\mathcal{K}} x_k^{(\infty)} - \sum_{k\in\mathcal{K}} (x_k^{(\infty)})^2 > 0$ holds when $\lambda \to \infty$, we have $\phi(\lambda) = -\sum_{k\in\mathcal{K}} x_k^{(\lambda)} + \lambda\sum_{k\in\mathcal{K}} x_k^{(\lambda)} - \lambda\sum_{k\in\mathcal{K}} x_k^{(\lambda)} \to +\infty$, contradicting with (68). Therefore, it can be concluded that $\sum_{k\in\mathcal{K}} x_k^{(\infty)} - \sum_{k\in\mathcal{K}} (x_k^{(\infty)})^2 = 0$ must hold when $\lambda \to \infty$. Consequently, $(\boldsymbol{x}^\infty, \boldsymbol{r}^\infty, \boldsymbol{y}^\infty)$ is feasible to (22a), i.e.,

$$\lim_{\lambda\to\infty} \phi(\lambda) = \mathcal{L}\left(\boldsymbol{x}^\infty, \infty\right) \geq P^\star \tag{71}$$

implying

$$\phi(\lambda) = P^\star, \lambda \to \infty. \tag{72}$$

Together with the fact that $\phi(\lambda)$ is a monotonic increasing function, it can be concluded that $\phi(\lambda)$ will obtain an upper bound at some λ_0^2, i.e., $\mathcal{L}(\boldsymbol{x}^\infty, \infty) = \phi(\lambda_0^2) = P^\star$. Then, we can obtain the same conclusion as (70), which indicates the strong Lagrangian duality holding for problem (22a). Therefore, problems (22a) and (23a) are equivalent when $\lambda \geq \lambda_0$ with λ_0 being a constant.

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